HuGaDB pipeline replay gallery

36 figures (nine sessions × four models)

AutomationForExoskeleton

April 6, 2026

Four pipeline replay variants. Each session page below arranges the same four plots in a 2×2 grid (top: binary path; bottom: multiclass path; left: SVM; right: LSTM).

Top row — binary locomotion (assist vs. not), then FSM

Binary SVM + FSM
Sliding-window IMU features feed a trained RBF SVM; the predicted binary class drives `RealtimeFsm` pseudo-real-time assist gating and the replay time series (vs. ground truth and IMU).

Binary LSTM + FSM
Same windowing and binary labels as the SVM path, but a trained LSTM outputs the binary class each step; FSM loop and plot layout match the SVM replay for direct comparison.

Bottom row — twelve activity classes, mapped to locomotion FSM

Multiclass ECOC SVM + FSM
ECOC SVM predicts twelve HuGaDB activity classes per window; class → locomotion mapping feeds the multiclass FSM and exo-command replay (with activity traces vs. ground truth where available).

Multiclass LSTM + FSM
Twelve-class LSTM replaces the ECOC head; same activity names, FSM mapping, and replay figure structure as the multiclass SVM pipeline.

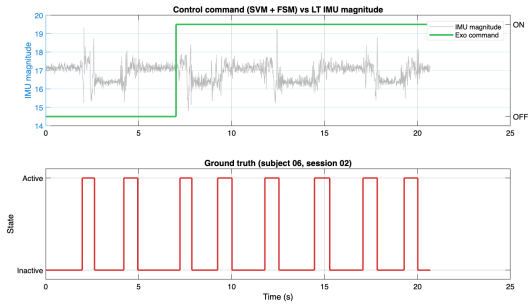
Abstract

This document collects every pseudo-real-time pipeline PNG for the nine HuGaDB `multi_activity_sequence` sessions used in `RunReplayGalleryBatch`. Each session folder under `results/figures/pipeline/` holds four plots: binary RBF SVM, binary LSTM, native twelve-class ECOC SVM, and native twelve-class LSTM, with the same FSM assist gating (binary path) or activity-to-locomotion mapping (multiclass path). Regenerate assets with `RunReplayGalleryBatch` from the MATLAB project root.

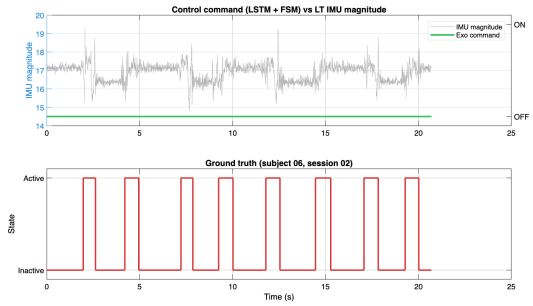
Held-out subjects 06 and 08 (three sessions each)

Subject 06, session 02

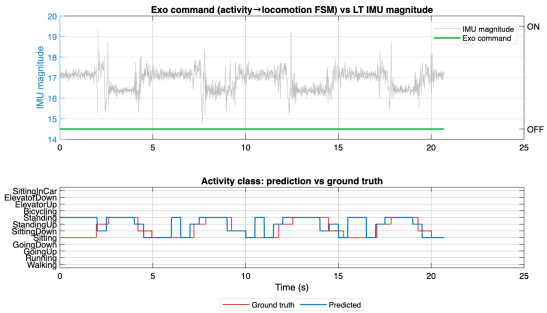
`subject06_session02` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



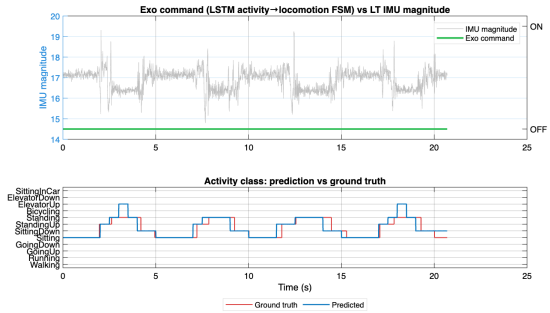
Binary SVM + FSM



Binary LSTM + FSM



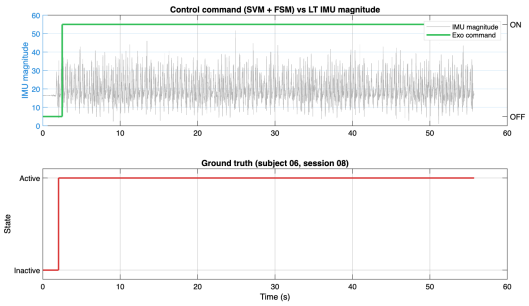
Multiclass ECOC SVM + FSM



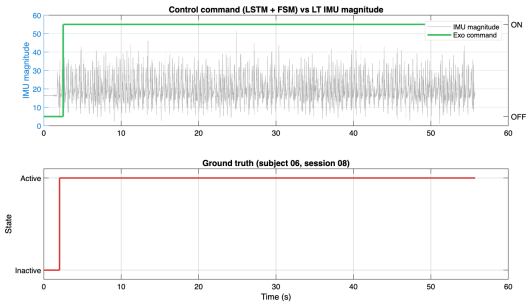
Multiclass LSTM + FSM

Subject 06, session 08

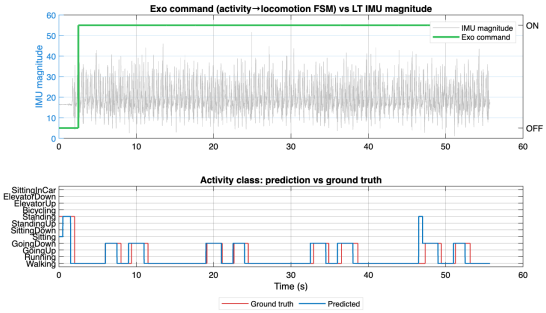
`subject06_session08` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



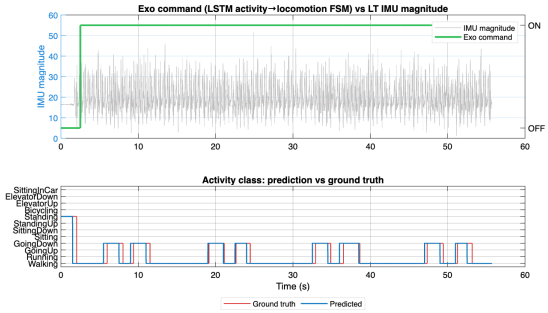
Binary SVM + FSM



Binary LSTM + FSM



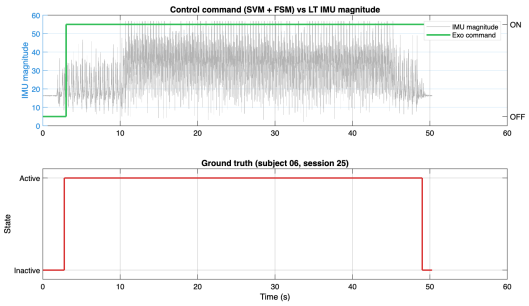
Multiclass ECOC SVM + FSM



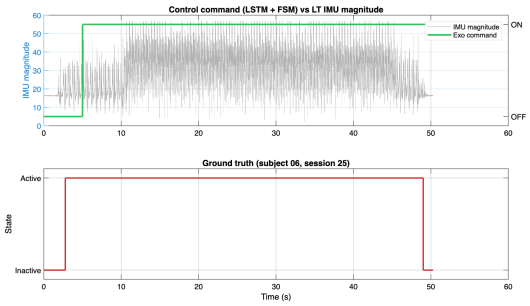
Multiclass LSTM + FSM

Subject 06, session 25

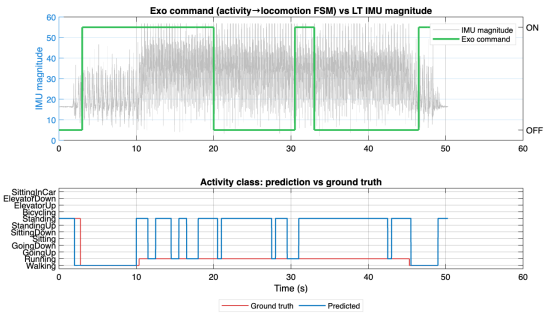
`subject06_session25` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



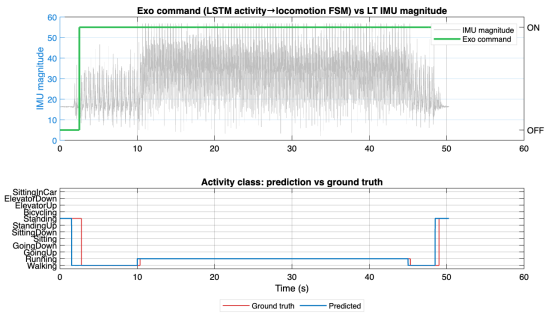
Binary SVM + FSM



Binary LSTM + FSM



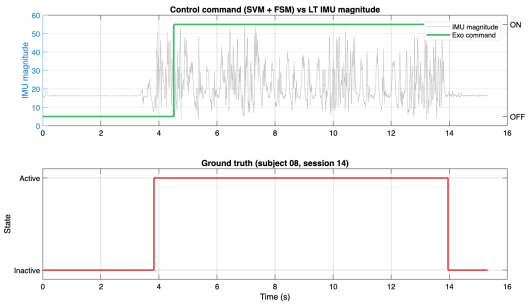
Multiclass ECOC SVM + FSM



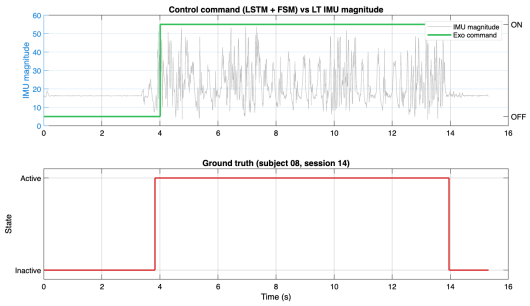
Multiclass LSTM + FSM

Subject 08, session 14

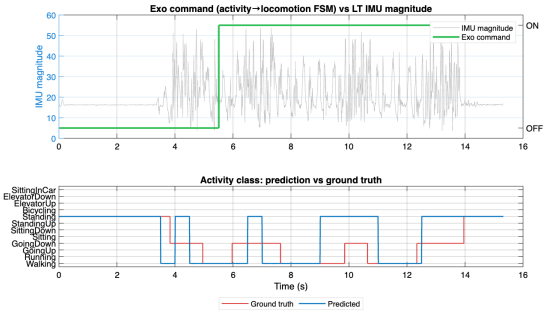
`subject08_session14` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



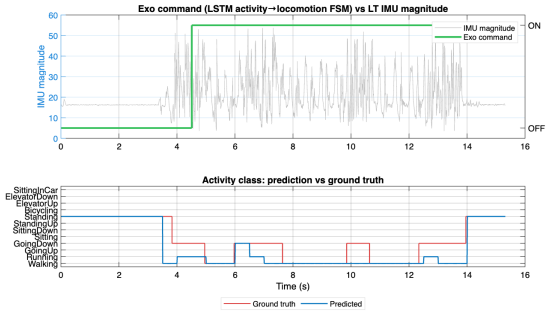
Binary SVM + FSM



Binary LSTM + FSM



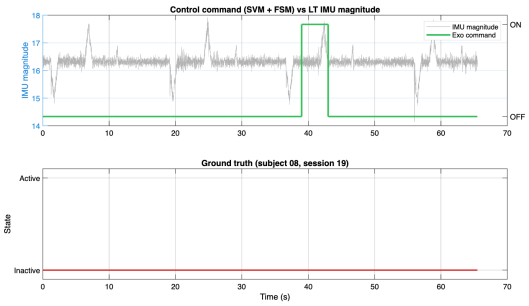
Multiclass ECOC SVM + FSM



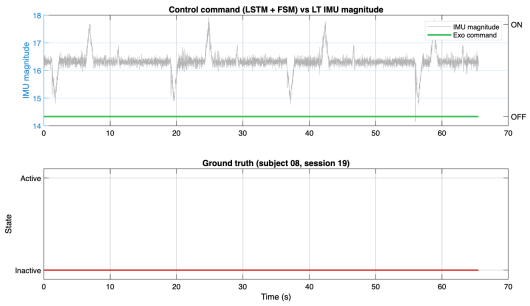
Multiclass LSTM + FSM

Subject 08, session 19

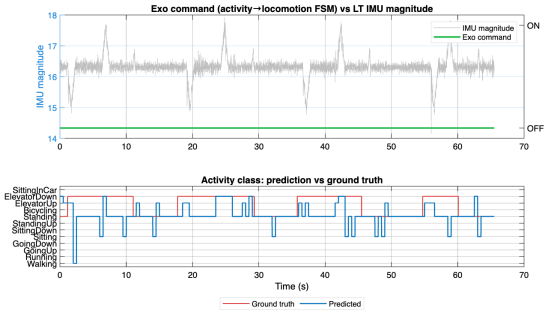
`subject08_session19` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



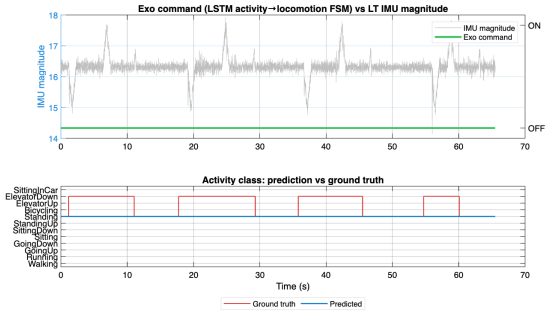
Binary SVM + FSM



Binary LSTM + FSM



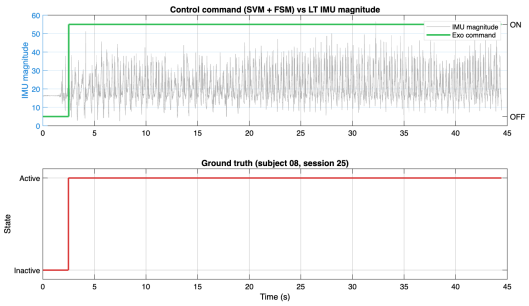
Multiclass ECOC SVM + FSM



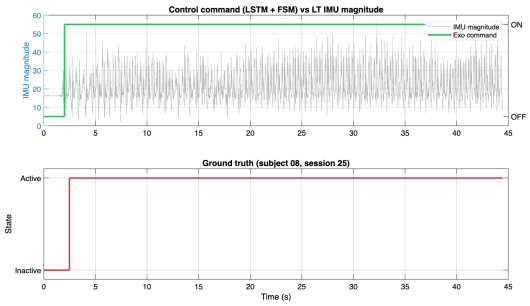
Multiclass LSTM + FSM

Subject 08, session 25

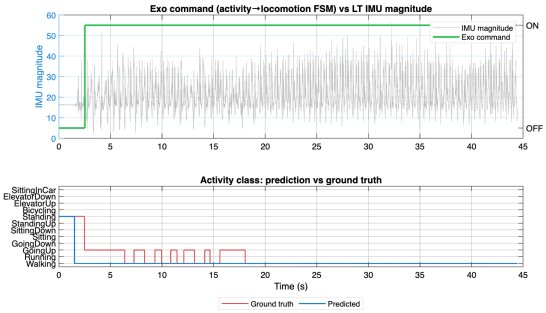
`subject08_session25` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



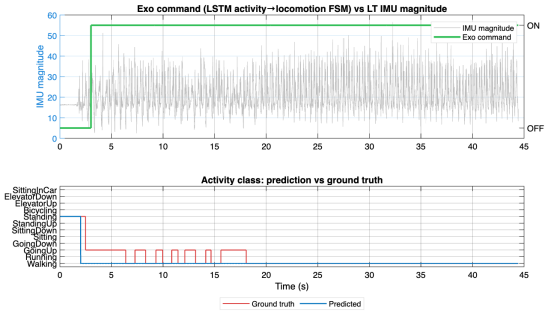
Binary SVM + FSM



Binary LSTM + FSM



Multiclass ECOC SVM + FSM

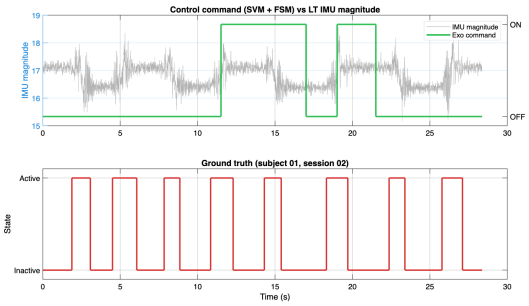


Multiclass LSTM + FSM

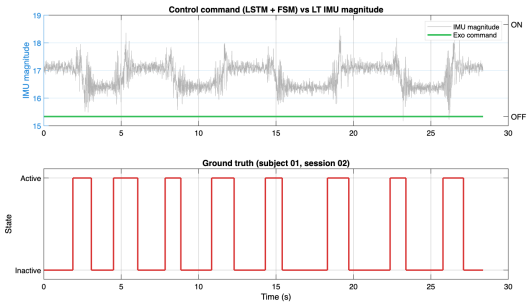
Additional subjects 01 and 02

Subject 01, session 02

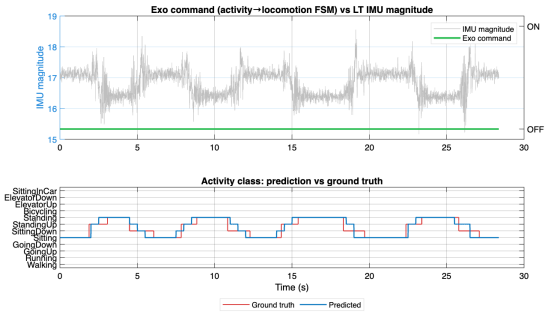
`subject01_session02` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



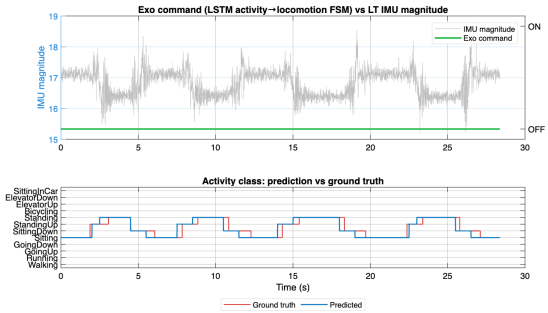
Binary SVM + FSM



Binary LSTM + FSM



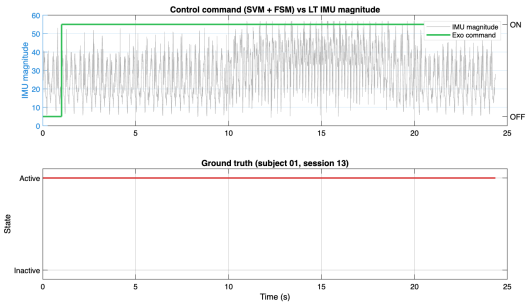
Multiclass ECOC SVM + FSM



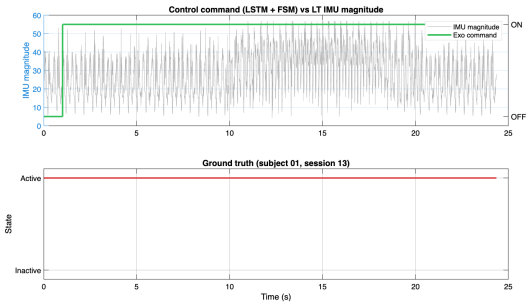
Multiclass LSTM + FSM

Subject 01, session 13

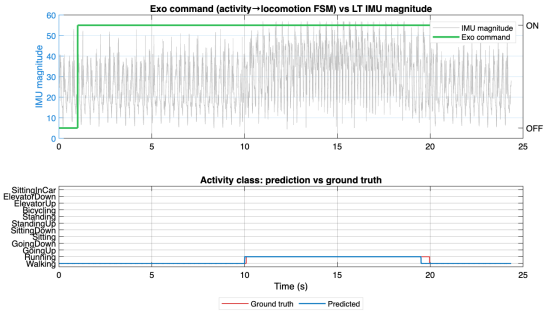
`subject01_session13` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



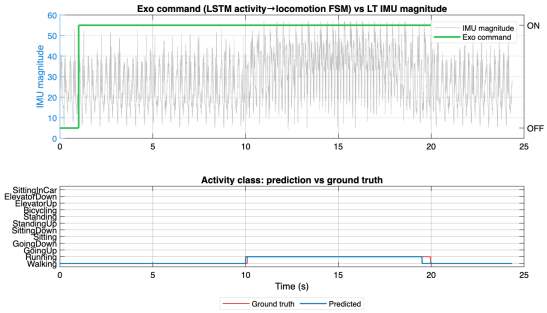
Binary SVM + FSM



Binary LSTM + FSM



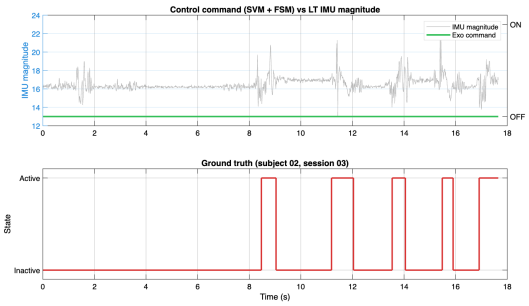
Multiclass ECOC SVM + FSM



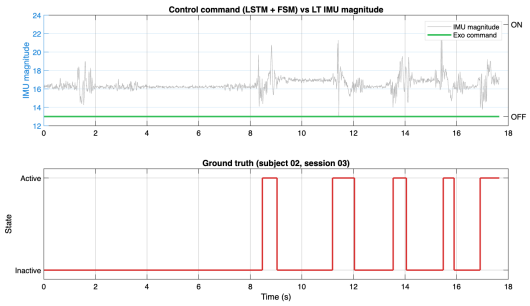
Multiclass LSTM + FSM

Subject 02, session 03

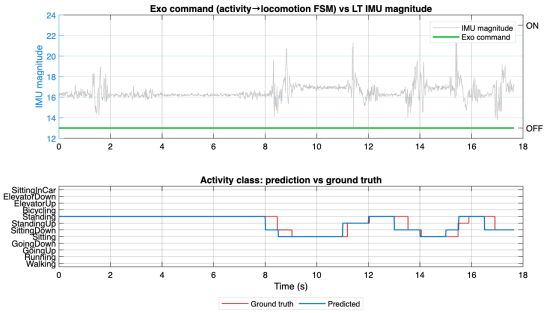
`subject02_session03` — binary SVM, binary LSTM, multiclass ECOC SVM, multiclass LSTM (same sliding-window + FSM structure as `RunReplayGalleryBatch`).



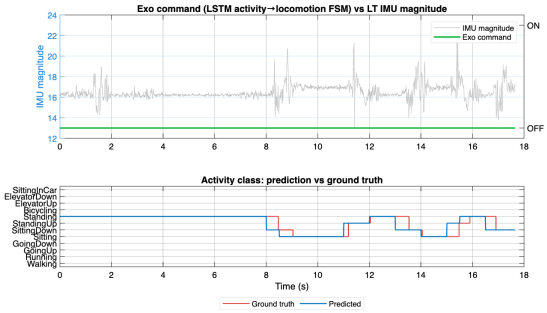
Binary SVM + FSM



Binary LSTM + FSM



Multiclass ECOC SVM + FSM



Multiclass LSTM + FSM